Two Lifted Actions on Oriented Configuration Spaces: Stabilization and Bredon Homology

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Abstract

Let C_2 act on $M = \mathbb{R}^r \oplus \sigma$, $r \geq 2$, by the reflection in the last coordinate. For $n \geq 2$ the induced action on the unordered configuration space $C_n(M)$ has two lifts to its orientation double cover $C_n^+(M)$. Adding a fixed point to a configuration defines a stabilization map u that is equivariant for each lift, whereas adding a free orbit defines a map v that exchanges the two lifts. We therefore treat both lifts together as one family $\mathcal{C}_{\mathrm{or}}(M)$, on which u and v are commuting equivariant self-maps. We prove that the Bredon homology of $\mathcal{C}_{\mathrm{or}}(M)$ with coefficients in the constant Mackey functor $\underline{\mathbb{Z}}$ is finitely generated over $\mathbb{Z}[u, v]$ in every degree, and over $\mathbb{Z}[u, v^2]$ for each lift separately. The proof follows the geometric fixed point method of Belmont, Quigley and Vogeli and rests on a description of the fixed points of both lifts. In degree zero we give a presentation of this Bredon homology as a bigraded $\mathbb{Z}[u, v]$-module. It has a cyclic direct summand with annihilator $(2, u^2)$, on which u has nonzero kernel for arbitrarily large numbers of points; for unordered configurations of M the map that adds a fixed point is injective in degree zero.

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1 Introduction

For a manifold N and $n \geq 2$, the oriented configuration space $C_n^+(N) = \mathrm{Conf}_n(N)/A_n$ is the double cover of the unordered configuration space $C_n(N) = \mathrm{Conf}_n(N)/\Sigma_n$ whose points are n-element subsets of N together with an ordering up to even permutations. When N is the interior of a manifold with boundary, adding a point near the boundary defines stabilization maps $C_n(N) \to C_{n+1}(N)$ and $C_n^+(N) \to C_{n+1}^+(N)$. Homological stability for the maps $C_n(N) \to C_{n+1}(N)$ goes back to McDuff [7] and Segal [13]. For connected N of dimension at least two, Randal-Williams [10] proved that they induce isomorphisms on $H_i(-;\mathbb{Z})$ for $2i \leq n$, and Palmer [9] proved the same for the maps $C_n^+(N) \to C_{n+1}^+(N)$ when $3i \leq n-5$; the precise statements are recalled in Section 4.

We study the oriented configuration spaces of a Euclidean space with a reflection from the point of view of equivariant homological stability. Let $C_2 = \{1, g\}$, let σ be its sign representation, and fix $r \geq 2$. We consider

$$M = \mathbb{R}^r \oplus \sigma, \qquad g(s,t) = (s,-t), \qquad F = M^{C_2} = \mathbb{R}^r \times \{0\}.$$

The diagonal action of g on $\mathrm{Conf}_n(M)$ descends to $C_n(M)$ and to $C_n^+(M)$. The action on $C_n(M)$ has a second lift to $C_n^+(M)$: if ν denotes the deck transformation of $C_n^+(M) \to C_n(M)$, then νg is again an involution. For $\epsilon \in \mathbb{Z}/2$ we write $C_n^{+,\epsilon}(M)$ for the space $C_n^+(M)$ with the action of $g_\epsilon = \nu^\epsilon g$, and we refer to ϵ as the lift.

In the equivariant setting a configuration can be enlarged by a whole orbit. For unordered configuration spaces of a manifold with an action of a finite group G, Belmont, Quigley and Vogeli [1] define, under suitable hypotheses, a stabilization map for each subgroup $H \leq G$, which adds an orbit of type G/H. They show that a single one of these maps does not in general give stability for Bredon homology, and they prove for Dedekind groups G that the Bredon homology of the disjoint union of all the unordered configuration spaces is finitely generated over the polynomial ring generated by these maps.

For C_2 acting on M there are two such maps: u adds a fixed point and v adds a free orbit $\{p, gp\}$. On oriented configurations u is equivariant for each of the two lifts. The map v is not: the new orbit is inserted as the ordered pair (p, gp), the reflection reverses this pair, and so $gv = \nu v g$. This makes v an equivariant map from $C_n^{+,\epsilon}(M)$ to $C_{n+2}^{+,\epsilon+1}(M)$, and we consider both lifts at once. Set

$$\mathcal{C}_{\mathrm{or}}(M) = \coprod_{\epsilon \in \mathbb{Z}/2} \mathcal{C}_{\mathrm{or}}^\epsilon(M), \qquad \mathcal{C}_{\mathrm{or}}^\epsilon(M) = \coprod_{n \geq 2} C_n^{+,\epsilon}(M).$$

We call $\mathcal{C}_{\mathrm{or}}^0(M)$ and $\mathcal{C}_{\mathrm{or}}^1(M)$ the two sectors. With the choices made in Section 2, u and v are commuting equivariant self-maps of $\mathcal{C}_{\mathrm{or}}(M)$,

$$u\colon C_n^{+,\epsilon}(M) \longrightarrow C_{n+1}^{+,\epsilon}(M), \qquad v\colon C_n^{+,\epsilon}(M) \longrightarrow C_{n+2}^{+,\epsilon+1}(M).$$

The homology of $\mathcal{C}_{\mathrm{or}}(M)$, in any homology theory for C_2-spaces, is then a module over

$$R = \mathbb{Z}[u, v],$$

bigraded by $(n, \epsilon) \in \mathbb{Z} \times \mathbb{Z}/2$ with $|u| = (1,0)$ and $|v| = (2,1)$. We use Bredon homology $H_*^{C_2}(-;\underline{\mathbb{Z}})$ with coefficients in the constant Mackey functor $\underline{\mathbb{Z}}$, evaluated at the orbit C_2/C_2; the conventions are recalled in Section 5.

Theorem 1.1. *For every $d \geq 0$, the R-module $H_d^{C_2}(\mathcal{C}_{\mathrm{or}}(M);\underline{\mathbb{Z}})$ is finitely generated.*

Since v exchanges the two sectors, the Bredon homology of a single sector is a module over the subring $\mathbb{Z}[u, v^2]$.

Corollary 1.2. *For every $d \geq 0$ and $\epsilon \in \mathbb{Z}/2$, the module $H_d^{C_2}(\mathcal{C}_{\mathrm{or}}^{\epsilon}(M);\underline{\mathbb{Z}})$ is finitely generated over $\mathbb{Z}[u, v^2]$.*

The proof of Theorem 1.1, given in Section 5, follows the method of [1]. Isotropy separation reduces the theorem to finite generation over R of the ordinary homology of the underlying space and of the fixed point space of $\mathcal{C}_{\mathrm{or}}(M)$. For the underlying space this follows from Palmer's stability theorem. The fixed point space is described as follows. A configuration $Q \in C_n(M)$ that is fixed by g consists of a points of F and b free orbits, with $n = a + 2b$; we say that Q has orbit type (a, b). The permutation of Q induced by g has sign $(-1)^b$, so the two points of $C_n^+(M)$ over Q are fixed by g_ϵ if $b \equiv \epsilon \pmod 2$ and are exchanged by g_ϵ otherwise. For $a + 2b \geq 2$, let $D_{a,b}$ be the space of fixed points of orbit type (a, b) in $C_{a+2b}^{+,\bar{b}}(M)$, where $\bar{b}$ is the class of b in $\mathbb{Z}/2$. We call $D_{a,b}$ a branch. The fixed point space of $\mathcal{C}_{\mathrm{or}}(M)$ is the disjoint union of the branches, and Theorem 3.3 gives homeomorphisms

$$D_{a,b} \cong \begin{cases} C_a^+(F) \times C_b(B), & a \geq 2, \\ F \times C_b(B) \times \{\pm 1\}, & a = 1, \\ C_b(B) \times \{\pm 1\}, & a = 0, \end{cases} \tag{1.1}$$

where $B = (M \setminus F)/C_2 \cong \mathbb{R}^r \times (0, \infty)$. We call the branches with $a \leq 1$ exceptional. If $a \leq 1$, the points of $Q \cap F$ admit no nontrivial permutation, the double cover is trivial over the configurations of orbit type (a, b), and the branch has two path components. The maps u and v send $D_{a,b}$ to $D_{a+1,b}$ and to $D_{a,b+1}$. In terms of (1.1), v is the stabilization map of the factor $C_b(B)$, and for $a \geq 2$ the map u is homotopic to the stabilization map of the factor $C_a^+(F)$. The map $u\colon D_{1,b} \to D_{2,b}$ sends both path components of $D_{1,b}$ into the path connected space $D_{2,b}$. Together with the theorems of Palmer and Randal-Williams, this gives finite generation of $H_d(\mathcal{C}_{\mathrm{or}}(M)^{C_2};\mathbb{Z})$ over R.

Section 6 computes the module of Theorem 1.1 in degree zero. Let x be the class of a point of $D_{2,0}$, let y be the class of a point in one of the two path components of $D_{0,1}$, and let δ be the difference of the classes of the two path components of $D_{0,1}$.

Theorem 1.3. *There is an isomorphism of bigraded R-modules*

$$H_0^{C_2}(\mathcal{C}_{\mathrm{or}}(M);\underline{\mathbb{Z}}) \cong \frac{R\langle x, y\rangle}{(vx - u^2 y,\ 2(vy - u^2 x))} \oplus \frac{R}{(2, u^2)}\langle \delta \rangle,$$

where $|x| = (2, 0)$ and $|y| = |\delta| = (2, 1)$.

The relation $vx = u^2 y$ holds because both sides are the class of the path connected branch $D_{2,1}$. The classes vy and $u^2 x$ come from two different path components of the fixed points of $C_4^{+,0}(M)$, namely a path component of $D_{0,2}$ and the branch $D_{4,0}$. In $H_0^{C_2}$ of a path connected C_2-space, the difference of the classes of two path components of the fixed points has order two (Lemma 6.1).

The summand generated by δ comes from the exceptional branches: $v^k\delta$ and $uv^k\delta$ are the differences of the classes of the two path components of $D_{0,k+1}$ and of $D_{1,k+1}$, and $u^2\delta = 0$ because u maps both path components of $D_{1,1}$ into $D_{2,1}$. So $uv^k\delta$ is a nonzero element of $H_0^{C_2}(C_{2k+3}^{+,\overline{k+1}}(M);\underline{\mathbb{Z}})$ in the kernel of u for every $k \geq 0$, and in each sector u fails to be injective on $H_0^{C_2}$ for infinitely many n. For the unordered configuration spaces of M the map induced by adding a fixed point is injective on $H_0^{C_2}$ for every n (Section 6.3).

Related work

Miller and Palmer [8] determine the stable homology of oriented configuration spaces by a scanning map into a double cover of a section space [8, Theorem A and Section 3.6]. They observe that the

stabilization map $C_n(N) \to C_{n+1}(N)$ has two lifts to the oriented configuration spaces, which differ by the deck transformation [8, Section 3.4]. Those are lifts of a map, whereas ϵ labels the lifts of the group action. For the free-orbit stabilization the two lifts in the sense of Miller and Palmer are v and νv, and νv satisfies the same relation $g(\nu v) = \nu(\nu v)g$ as v.

Rourke and Sanderson [12] prove equivariant versions of the classical theorems relating configuration spaces to iterated loop spaces. Randal-Williams [11] studies the spaces $C_n(N)$ for all n simultaneously: their one-point compactifications form a commutative monoid under superposition of configurations, and this monoid is described by a homotopy pushout square [11, Theorem 1.1]. For a finite group G and a real G-representation V, Dugger and Hazel [4] study the $RO(G)$-graded Bredon cohomology of the ordered configuration spaces $\mathrm{Conf}_n(V)$ with constant integral coefficients, for a fixed number of points. When V contains a trivial summand, they show that it is free over the cohomology of a point and describe the ring structure by generators and relations.

2 The two lifts and the stabilization maps

2.1 Oriented configuration spaces

For a space X and $n \geq 0$, let

$$\mathrm{Conf}_n(X) = \{(x_1, \ldots, x_n) \in X^n : x_i \neq x_j \text{ for } i \neq j\}$$

be the space of ordered configurations. The symmetric group Σ_n acts freely on it by permuting the coordinates, and we put

$$C_n(X) = \mathrm{Conf}_n(X)/\Sigma_n, \qquad C_n^+(X) = \mathrm{Conf}_n(X)/A_n \quad (n \geq 2).$$

Here $C_0(X)$ is a point and $C_1(X) = X$. A point of $C_n(X)$ is an n-element subset $Q \subset X$. A point of $C_n^+(X)$ is written $[x_1, \ldots, x_n]$, the class of an ordered configuration, or (Q, o), where o is an orientation of Q, that is, an ordering of Q up to even permutations. The projection $C_n^+(X) \to C_n(X)$ is a two-sheeted covering. Its deck transformation ν is induced by any odd permutation of the coordinates, and $\nu(Q, o) = (Q, -o)$.

If N is a connected manifold of dimension at least two, then $\mathrm{Conf}_n(N)$ is path connected, and hence so are $C_n(N)$ and $C_n^+(N)$.

2.2 The two lifts

Throughout, $r \geq 2$ is fixed, σ is the sign representation of $C_2 = \{1, g\}$, and

$$M = \mathbb{R}^r \oplus \sigma.$$

Points of M are written (s, t) with $s = (s_1, \ldots, s_r) \in \mathbb{R}^r$ and $t \in \mathbb{R}$, so that $g(s, t) = (s, -t)$ and the fixed point space is $F = \mathbb{R}^r \times \{0\}$. The diagonal action of g on $\mathrm{Conf}_n(M)$ commutes with the action of Σ_n. It induces involutions of $C_n(M)$ and of $C_n^+(M)$, both again denoted by g, and on $C_n^+(M)$ we have $g\nu = \nu g$.

Definition 2.1. Let $n \geq 2$ and $\epsilon \in \mathbb{Z}/2$. We write $g_\epsilon = \nu^\epsilon g$, and $C_n^{+,\epsilon}(M)$ denotes the space $C_n^+(M)$ with the C_2-action in which g acts by g_ϵ.

Proposition 2.2. *The maps g_0 and g_1 are involutions of $C_n^+(M)$, and they are the only self-maps of $C_n^+(M)$ that cover the involution g of $C_n(M)$.*

Proof. Since g and ν commute and both square to the identity, $g_\epsilon^2 = \nu^{2\epsilon} g^2 = 1$. Both g_0 and g_1 cover g because ν covers the identity. Let γ be any map of $C_n^+(M)$ covering g. For each z, the point $\gamma(z)$ is either $g(z)$ or $\nu g(z)$, and the set of z with $\gamma(z) = g(z)$ is open and closed because $C_n^+(M) \to C_n(M)$ is a covering. As $C_n^+(M)$ is connected, γ is g_0 or g_1. □

We collect the spaces $C_n^{+,\epsilon}(M)$ into the C_2-spaces

$$\mathcal{C}_{\mathrm{or}}^{\epsilon}(M) = \coprod_{n\geq 2} C_n^{+,\epsilon}(M), \qquad \mathcal{C}_{\mathrm{or}}(M) = \mathcal{C}_{\mathrm{or}}^0(M) \amalg \mathcal{C}_{\mathrm{or}}^1(M),$$

and call $\mathcal{C}_{\mathrm{or}}^0(M)$ and $\mathcal{C}_{\mathrm{or}}^1(M)$ the sectors of $\mathcal{C}_{\mathrm{or}}(M)$. The summands $C_n^{+,0}(M)$ and $C_n^{+,1}(M)$ have the same underlying space and different C_2-actions.

2.3 Stabilization maps

Define embeddings $e_u, e_v \colon M \to M$ by

$$e_u(s_1, s_2, \ldots, s_r, t) = (1 - e^{-s_1}, s_2, \ldots, s_r, t), \qquad e_v(s,t) = (s, \tanh t),$$

and let

$$q = (2, 0, \ldots, 0; 0) \in F, \qquad p = (0, \ldots, 0; 2) \in M.$$

Both embeddings are C_2-equivariant and they commute. The image of e_u lies in $\{s_1 < 1\}$ and that of e_v in $\{|t| < 1\}$, so $q \notin e_u(M)$ and $p, gp \notin e_v(M)$. Moreover

$$e_u(p) = p, \qquad e_u(gp) = gp, \qquad e_v(q) = q. \tag{2.1}$$

On ordered configurations we define

$$\tilde{u}(x_1, \ldots, x_n) = (e_u x_1, \ldots, e_u x_n, q), \qquad \tilde{v}(x_1, \ldots, x_n) = (e_v x_1, \ldots, e_v x_n, p, gp).$$

These maps are equivariant with respect to the inclusions $\Sigma_n \subset \Sigma_{n+1}$ and $\Sigma_n \subset \Sigma_{n+2}$ as the permutations of the first n coordinates. They induce maps

$$u \colon C_n^+(M) \longrightarrow C_{n+1}^+(M), \qquad v \colon C_n^+(M) \longrightarrow C_{n+2}^+(M)$$

for every $n \geq 2$.

Lemma 2.3. *The maps u and v satisfy*

$$u\nu = \nu u, \qquad v\nu = \nu v, \qquad gu = ug, \qquad gv = \nu v g.$$

Proof. An odd permutation of the first n coordinates induces ν on the source and on the target of u and of v, which gives the first two relations. Since e_u is equivariant and $gq = q$, we have $g\tilde{u} = \tilde{u}g$. For v,

$$g\tilde{v}(x_1, \ldots, x_n) = (e_v g x_1, \ldots, e_v g x_n, gp, p),$$

which is $\tilde{v}(gx_1, \ldots, gx_n)$ with its last two coordinates transposed. This gives $gv = \nu v g$. □

Proposition 2.4. *Let $n \geq 2$ and $\epsilon \in \mathbb{Z}/2$.*

(a) *The maps $u \colon C_n^{+,\epsilon}(M) \to C_{n+1}^{+,\epsilon}(M)$ and $v \colon C_n^{+,\epsilon}(M) \to C_{n+2}^{+,\epsilon+1}(M)$ are C_2-equivariant.*

(b) *$uv = vu$ as maps $C_n^+(M) \to C_{n+3}^+(M)$.*

Proof. (a) By Lemma 2.3,

$$g_\epsilon u = \nu^\epsilon g u = \nu^\epsilon u g = u g_\epsilon, \qquad g_{\epsilon+1} v = \nu^{\epsilon+1} g v = \nu^{\epsilon+2} v g = v \nu^\epsilon g = v g_\epsilon.$$

(b) Let $x = (x_1, \dots, x_n) \in \mathrm{Conf}_n(M)$ and put $x'_i = e_u e_v x_i = e_v e_u x_i$. By (2.1),

$$\tilde{u}\tilde{v}(x) = (x'_1, \dots, x'_n, p, gp, q), \qquad \tilde{v}\tilde{u}(x) = (x'_1, \dots, x'_n, q, p, gp).$$

The two differ by a cyclic permutation of the last three coordinates, which is even. □

Inserting the new orbit in the other order, as (gp, p), replaces v by νv. This map satisfies $g(\nu v) = \nu(\nu v)g$, so it exchanges the two lifts in the same way as v.

Let $R = \mathbb{Z}[u, v]$. By Proposition 2.4, u and v are commuting C_2-equivariant self-maps of $\mathcal{C}_{\mathrm{or}}(M)$, so every functor h from C_2-spaces to abelian groups makes $h(\mathcal{C}_{\mathrm{or}}(M))$ an R-module, and natural transformations between such functors induce R-linear maps. If h takes disjoint unions to direct sums, then

$$h(\mathcal{C}_{\mathrm{or}}(M)) = \bigoplus_{n \geq 2,\ \epsilon \in \mathbb{Z}/2} h(C_n^{+,\epsilon}(M))$$

is bigraded by (n, ϵ), and u and v have bidegrees $(1, 0)$ and $(2, 1)$. The same holds for functors of the fixed point space $\mathcal{C}_{\mathrm{or}}(M)^{C_2}$ and of the underlying space of $\mathcal{C}_{\mathrm{or}}(M)$, on which u and v also act.

3 Fixed points

3.1 Orbit types and parity

A configuration $Q \in C_n(M)$ is fixed by g if and only if it is a union of C_2-orbits; we call such a configuration invariant. It consists of a points of F and b free orbits $\{m, gm\}$, where $a + 2b = n$, and we say that Q has *orbit type* (a, b).

Proposition 3.1. *Let $Q \in C_n(M)^{C_2}$ have orbit type (a, b), and let o be an orientation of Q. Then*

$$g_\epsilon(Q, o) = (Q, (-1)^{b+\epsilon} o).$$

Hence both points of $C_n^{+,\epsilon}(M)$ over Q are fixed if $b \equiv \epsilon \pmod 2$, and neither is fixed otherwise.

Proof. The permutation of Q induced by g fixes the a points in F and transposes the two points of each free orbit, so its sign is $(-1)^b$ and $g(Q, o) = (Q, (-1)^b o)$. Applying ν^ϵ multiplies the orientation by $(-1)^\epsilon$. □

For each invariant configuration Q, exactly one of the two lifts has fixed points over Q, namely the one with $\epsilon = \bar{b}$, where $\bar{b}$ denotes the class of b in $\mathbb{Z}/2$.

Definition 3.2. For $a, b \geq 0$ with $a + 2b \geq 2$, the *branch* $D_{a,b}$ is the subspace of the fixed point space of $C_{a+2b}^{+,\bar{b}}(M)$ consisting of the points (Q, o) with Q of orbit type (a, b). The branches with $a \leq 1$ are called *exceptional*.

The orbit type is locally constant on $C_n(M)^{C_2}$. To see this, choose pairwise disjoint open balls around the points of an invariant configuration Q such that g permutes the balls. A configuration close to Q has exactly one point in each ball. If it is invariant, its point in a ball centered on F is fixed, and its points in the two balls around a free orbit form a free orbit. It follows that the branches are open and closed in the fixed point space, and Proposition 3.1 gives

$$C_n^{+,\epsilon}(M)^{C_2} = \coprod_{\substack{a+2b=n \\ b \equiv \epsilon\ (2)}} D_{a,b}, \qquad \mathcal{C}_{\mathrm{or}}(M)^{C_2} = \coprod_{a+2b \geq 2} D_{a,b}. \tag{3.1}$$

3.2 The branches

Let $M_{>0} = \mathbb{R}^r \times (0,\infty)$ and let

$$B = (M \setminus F)/C_2$$

be the space of free orbits. Every free orbit contains exactly one point of $M_{>0}$, which we call its *positive representative*, and the quotient map restricts to a homeomorphism $M_{>0} \to B$. We use this homeomorphism to identify B with $M_{>0} \cong \mathbb{R}^{r+1}$.

Theorem 3.3. *Let $a, b \geq 0$ with $a+2b \geq 2$. For a configuration $\{\beta_1, \dots, \beta_b\} \in C_b(B)$, let $m_i \in M_{>0}$ be the positive representative of β_i. The following maps are well-defined homeomorphisms.*

(a) *For $a \geq 2$,*

$$\Theta_{a,b} \colon C_a^+(F) \times C_b(B) \longrightarrow D_{a,b}, \qquad ([f_1, \dots, f_a], \{\beta_i\}) \longmapsto [f_1, \dots, f_a, m_1, gm_1, \dots, m_b, gm_b].$$

(b) *For $a = 1$,*

$$\Theta_{1,b} \colon F \times C_b(B) \times \{\pm 1\} \longrightarrow D_{1,b},$$

which sends $(f, \{\beta_i\}, +1)$ to $[f, m_1, gm_1, \dots, m_b, gm_b]$ and $(f, \{\beta_i\}, -1)$ to the image of this point under ν.

(c) *For $a = 0$,*

$$\Theta_{0,b} \colon C_b(B) \times \{\pm 1\} \longrightarrow D_{0,b},$$

which sends $(\{\beta_i\}, +1)$ to $[m_1, gm_1, \dots, m_b, gm_b]$ and $(\{\beta_i\}, -1)$ to the image of this point under ν.

Consequently $D_{a,b}$ is path connected for $a \geq 2$ and has two path components for $a \leq 1$.

For $a \leq 1$ we write $D_{a,b}^+$ and $D_{a,b}^-$ for the images under $\Theta_{a,b}$ of the parts with last coordinate $+1$ and -1.

Proof. The formulas do not depend on the numbering of the orbits β_i, because moving the pair (m_i, gm_i) past another pair is an even permutation, and for $a \geq 2$ they do not depend on the representative of $[f_1, \dots, f_a]$, because an even permutation of the f_j is an even permutation of the whole list. The maps are continuous since the positive representative of a free orbit depends continuously on the orbit.

Let $n = a + 2b$ and let $I_{a,b} \subset C_n(M)^{C_2}$ be the subspace of configurations of orbit type (a, b). Sending Q to the pair $(Q \cap F, (Q \setminus F)/C_2)$ is a homeomorphism $I_{a,b} \to C_a(F) \times C_b(B)$. By Proposition 3.1, $D_{a,b}$ is the full preimage of $I_{a,b}$ in $C_n^+(M)$, so it is a two-sheeted covering of $I_{a,b}$ with deck transformation ν. The source of $\Theta_{a,b}$ is a two-sheeted covering of $C_a(F) \times C_b(B)$: for $a \geq 2$ through the projection $C_a^+(F) \to C_a(F)$, and for $a \leq 1$ by forgetting the sign. The map $\Theta_{a,b}$ covers the homeomorphism above, and it commutes with the deck transformations, since an odd permutation of the f_j, or a change of the sign, changes the image by ν. A map of two-sheeted coverings over a homeomorphism of the bases that commutes with the deck transformations is a homeomorphism.

The spaces $F \cong \mathbb{R}^r$, $C_b(B)$ and $C_a^+(F)$ for $a \geq 2$ are path connected because $r \geq 2$. This gives the number of path components of $D_{a,b}$. □

Example 3.4. By (3.1) and Theorem 3.3, the fixed point spaces for $2 \le n \le 5$ are

$$\begin{aligned}
C_2^{+,0}(M)^{C_2} &= D_{2,0} \cong C_2^+(F), & C_2^{+,1}(M)^{C_2} &= D_{0,1} \cong B \times \{\pm 1\},\\
C_3^{+,0}(M)^{C_2} &= D_{3,0} \cong C_3^+(F), & C_3^{+,1}(M)^{C_2} &= D_{1,1} \cong F \times B \times \{\pm 1\},\\
C_4^{+,0}(M)^{C_2} &= D_{4,0} \amalg D_{0,2}, & C_4^{+,1}(M)^{C_2} &= D_{2,1} \cong C_2^+(F) \times B,\\
C_5^{+,0}(M)^{C_2} &= D_{5,0} \amalg D_{1,2}, & C_5^{+,1}(M)^{C_2} &= D_{3,1} \cong C_3^+(F) \times B,
\end{aligned}$$

where $D_{0,2} \cong C_2(B) \times \{\pm 1\}$ and $D_{1,2} \cong F \times C_2(B) \times \{\pm 1\}$.

3.3 Stabilization on the branches

The embeddings e_u and e_v preserve F and $M_{>0}$, and e_v is the identity on F. Let $\bar{e}_u, \bar{e}_v \colon B \to B$ be the induced embeddings of the orbit space and let $\bar{p} \in B$ be the orbit of p. Under the identification $B = M_{>0}$, the maps $\bar{e}_u$ and $\bar{e}_v$ are the restrictions of e_u and e_v to $M_{>0}$, and $\bar{p} = p$. For an embedding e we write $C_b(e)$ for the induced map of configuration spaces. Define

$$\begin{aligned}
\mathrm{st}_B \colon C_b(B) &\longrightarrow C_{b+1}(B), & Z &\longmapsto \bar{e}_v(Z) \cup \{\bar{p}\},\\
\mathrm{st}_F \colon C_a^+(F) &\longrightarrow C_{a+1}^+(F) \quad (a \ge 2), & [f_1, \dots, f_a] &\longmapsto [e_u f_1, \dots, e_u f_a, q],\\
\lambda \colon F \times \{\pm 1\} &\longrightarrow C_2^+(F), & (f, +1) &\longmapsto [e_u f, q], \quad (f, -1) \longmapsto [q, e_u f].
\end{aligned}$$

Proposition 3.5. *The maps u and v send $D_{a,b}$ into $D_{a+1,b}$ and $D_{a,b+1}$, respectively. In terms of the homeomorphisms of Theorem 3.3 they are given as follows.*

(a) *For all a, the map $v \colon D_{a,b} \to D_{a,b+1}$ is st_B on the factor $C_b(B)$ and the identity on the other factors.*

(b) *The map $u \colon D_{a,b} \to D_{a+1,b}$ is*

$$\begin{aligned}
(\xi, Z) &\longmapsto (\mathrm{st}_F(\xi), C_b(\bar{e}_u)(Z)) && \text{for } a \ge 2,\\
(f, Z, \eta) &\longmapsto (\lambda(f, \eta), C_b(\bar{e}_u)(Z)) && \text{for } a = 1,\\
(Z, \eta) &\longmapsto (q, C_b(\bar{e}_u)(Z), \eta) && \text{for } a = 0.
\end{aligned}$$

(c) *The map $C_b(\bar{e}_u)$ is homotopic to the identity of $C_b(B)$. For $a \ge 2$ the map $u \colon D_{a,b} \to D_{a+1,b}$ therefore corresponds to a map homotopic to $\mathrm{st}_F \times \mathrm{id}$.*

Proof. (a) Let $m_1, \dots, m_b \in M_{>0}$. The map v applies e_v to all points and appends (p, gp). It fixes the points of F, the points $e_v m_i$ are the positive representatives of the orbits $\bar{e}_v(\beta_i)$, and p is the positive representative of $\bar{p}$. So v sends $[\dots, m_1, gm_1, \dots, m_b, gm_b]$ to $[\dots, e_v m_1, g e_v m_1, \dots, e_v m_b, g e_v m_b, p, gp]$, which is the image under $\Theta_{a,b+1}$ of the point with the same coordinates in the other factors and with $\mathrm{st}_B(\{\beta_i\})$ in $C_{b+1}(B)$. Since v commutes with ν, this also covers the points with sign -1.

(b) The map u applies e_u to all points and appends q at the end of the list. Moving q to the left across the $2b$ points $e_u m_1, g e_u m_1, \dots, e_u m_b, g e_u m_b$ is an even permutation. For $a \ge 2$ this gives

$$u[f_1, \dots, f_a, m_1, gm_1, \dots] = [e_u f_1, \dots, e_u f_a, q, e_u m_1, g e_u m_1, \dots],$$

which is the first formula. For $a = 0$ the point q becomes the single point in F, placed first, and the sign is unchanged because $u\nu = \nu u$. For $a = 1$ we obtain $[e_u f, q, e_u m_1, g e_u m_1, \dots]$ for $\eta = +1$ and

its image under ν, which is $[q, e_u f, e_u m_1, g e_u m_1, \dots]$, for $\eta = -1$. These are the images under $\Theta_{2,b}$ of the points with first coordinate $\lambda(f, \eta)$.

(c) For $0 \le \tau \le 1$ the function $s_1 \mapsto (1-\tau)(1-e^{-s_1}) + \tau s_1$ has positive derivative. Applying it to the first coordinate defines an isotopy of embeddings $B \to B$ from $\bar{e}_u$ to the identity, and hence a homotopy from $C_b(\bar{e}_u)$ to the identity. □

By part (b), u maps $D^{\pm}_{0,b}$ into $D^{\pm}_{1,b}$, and it maps both path components of $D_{1,b}$ into the path connected branch $D_{2,b}$: the points $\lambda(f, +1)$ and $\lambda(f, -1)$ are the two orientations of the subset $\{e_u f, q\}$ of F.

4 Finite generation in ordinary homology

In this section homology is singular homology with integer coefficients unless other coefficients are indicated.

4.1 Two stability theorems

Theorem 4.1 (Palmer [9, Main Theorem])**.** *Let N be the interior of a connected manifold with nonempty boundary, of dimension at least two. The stabilization map $C^+_a(N) \to C^+_{a+1}(N)$ induces an isomorphism on H_i for $i \le (a-5)/3$.*

Theorem 4.2 (Randal-Williams [10, Theorem A])**.** *Let N be the interior of a connected compact manifold with nonempty boundary, of dimension at least two. The stabilization map $C_b(N) \to C_{b+1}(N)$ induces an isomorphism on H_i for $2i \le b$.*

Remark 4.3. In both theorems N is the interior of a manifold with boundary $\bar{N}$. The stabilization map pushes a configuration away from a chosen point of $\partial\bar{N}$ by a self-embedding of N supported near the boundary, and then adds a new point in the vacated region; for oriented configurations the new point is placed last in the ordering. See [9, Definition 2.3] and [10, Section 4]. The homotopy class of this map depends only on the component of $\partial\bar{N}$ containing the chosen point. We apply the theorems to st_F, to the map underlying u, and to st_B, which are homotopic to maps of this kind for the following reasons.

For st_F and u the manifold is $N = \mathbb{R}^m$ with $m = r$ or $m = r+1$. Take $\bar{N} = (-\infty, \infty] \times \mathbb{R}^{m-1}$ and the boundary point $(\infty, 0, \dots, 0)$, and let $k\colon (-\infty, \infty] \to (-\infty, 2]$ be an increasing homeomorphism that is the identity on $(-\infty, 0]$. Applying k to the first coordinate of every point of a configuration and adding q as the last point is a stabilization map as in [9]. For $0 \le \tau \le 1$ the function $(1-\tau)(1-e^{-s_1}) + \tau k(s_1)$ is an increasing embedding of $\mathbb{R}$ into $(-\infty, 2)$. Applying it to the first coordinate and adding q as the last point gives a homotopy from st_F, respectively u, to this map.

For st_B we use on $B = \mathbb{R}^r \times (0, \infty)$ the coordinates (s, y) with $y = 1/t$. In these coordinates B is the interior of the closed half-space $\mathbb{R}^r \times [0, \infty)$, whose one-point compactification is a closed ball, the point $\bar{p}$ is $(0, 1/2)$, and $\bar{e}_v(s, y) = (s, \coth(1/y))$. Choose smooth functions with compact support $\beta\colon \mathbb{R}^r \to [0, 1]$ and $\chi\colon [0, \infty) \to [0, 1]$ such that $\beta(0) = 1$, $\chi = 1$ near 0 and $\chi' > -2$. Then

$$e(s, y) = \big(s,\ y + \tfrac{1}{2}\beta(s)\chi(y)\big)$$

is a self-embedding of the closed half-space with compact support. It is isotopic to the identity and moves the boundary point 0 to $\bar{p}$, so after rescaling the coordinates it is an embedding of the kind used in [10, Section 4], and the associated stabilization map is $Z \mapsto e(Z) \cup \{\bar{p}\}$. For $0 \le \tau \le 1$ consider

$$(s, y) \longmapsto \Big(s,\ (1-\tau)\coth(1/y) + \tau\big(y + \tfrac{1}{2}\beta(s)\chi(y)\big)\Big).$$

The last coordinate is increasing in y, so this is an embedding of B into itself. Its image does not contain $\bar{p}$, since for $s = 0$ and $y > 0$ both $\coth(1/y)$ and $y + \frac{1}{2}\chi(y)$ are greater than $1/2$. Applying these embeddings to Z and adding $\bar{p}$ gives a homotopy from st_B to the stabilization map defined by e.

4.2 Stability on the branches

Lemma 4.4. *The homology groups of $C_n^+(M)$ for $n \geq 2$, and of the branches $D_{a,b}$, are finitely generated in every degree.*

Proof. Let $m \geq 2$. The homology of $\mathrm{Conf}_n(\mathbb{R}^m)$ is finitely generated in every degree [3]. If a finite group G acts freely on a space X such that $X \to X/G$ is a covering, there is a spectral sequence

$$E^2_{i,j} = H_i(G; H_j(X)) \Longrightarrow H_{i+j}(X/G); \tag{4.1}$$

see [2, Chapter VII, Section 7]. The homology of a finite group with coefficients in a finitely generated module is finitely generated, so if $H_j(X)$ is finitely generated for all j, the same holds for $H_j(X/G)$. Applied to the actions of A_n and Σ_n on $\mathrm{Conf}_n(\mathbb{R}^m)$, this shows that $C_n^+(\mathbb{R}^m)$ and $C_n(\mathbb{R}^m)$ have finitely generated homology in every degree. The first claim follows with $m = r + 1$, since the space underlying $C_n^+(M)$ is $C_n^+(\mathbb{R}^{r+1})$. By Theorem 3.3, each path component of $D_{a,b}$ is a product of spaces of the form $C_a^+(\mathbb{R}^r)$, $\mathbb{R}^r$ and $C_b(\mathbb{R}^{r+1})$, and the second claim follows from the Künneth theorem. □

Proposition 4.5. *Let $d \geq 0$ and let $a, b \geq 0$ with $a + 2b \geq 2$.*

(a) *If $a \geq 3d + 5$, then $u_*\colon H_d(D_{a,b}) \to H_d(D_{a+1,b})$ is an isomorphism.*

(b) *If $b \geq 2d$, then $v_*\colon H_d(D_{a,b}) \to H_d(D_{a,b+1})$ is an isomorphism.*

Proof. We use the Künneth sequence

$$0 \to \bigoplus_{i+j=d} H_i(X) \otimes H_j(Y) \to H_d(X \times Y) \to \bigoplus_{i+j=d-1} \mathrm{Tor}\big(H_i(X), H_j(Y)\big) \to 0, \tag{4.2}$$

which is natural in X and Y. If a map $X \to X'$ induces isomorphisms on H_i for all $i \leq d$, then by (4.2) and the five lemma its product with the identity of Y induces an isomorphism on H_d.

(a) Here $a \geq 2$, and by Proposition 3.5(c) the map $u\colon D_{a,b} \to D_{a+1,b}$ corresponds to a map homotopic to $\mathrm{st}_F \times \mathrm{id}$ on $C_a^+(F) \times C_b(B)$. For $i \leq d$ we have $i \leq (a-5)/3$, so st_F induces an isomorphism on H_i by Theorem 4.1 and Remark 4.3.

(b) By Proposition 3.5(a), v corresponds to the product of $\mathrm{st}_B\colon C_b(B) \to C_{b+1}(B)$ with the identity of $C_a^+(F)$, of $F \times \{\pm 1\}$ or of $\{\pm 1\}$, according to whether $a \geq 2$, $a = 1$ or $a = 0$. For $i \leq d$ we have $2i \leq b$, so st_B induces an isomorphism on H_i by Theorem 4.2 and Remark 4.3. □

4.3 Finite generation

Recall from Section 2.3 that the homology of $\mathcal{C}_{\mathrm{or}}(M)^{C_2}$ and the homology of the underlying space of $\mathcal{C}_{\mathrm{or}}(M)$ are R-modules. By (3.1),

$$H_d(\mathcal{C}_{\mathrm{or}}(M)^{C_2}) = \bigoplus_{a+2b \geq 2} H_d(D_{a,b}).$$

Proposition 4.6. *Let $d \geq 0$.*

(a) *The R-module $H_d(\mathcal{C}_{\mathrm{or}}(M)^{C_2})$ is finitely generated; it is generated by the groups $H_d(D_{a,b})$ with $a \le 3d+5$ and $b \le \max\{2d,1\}$.*

(b) *The homology $H_d(\mathcal{C}_{\mathrm{or}}(M))$ of the underlying space is a finitely generated R-module; it is generated over $\mathbb{Z}[u]$ by the groups $H_d(C_n^{+,\epsilon}(M))$ with $n \le 3d+5$.*

Proof. (a) Put $a_d = 3d+5$ and $b_d = \max\{2d,1\}$, and let $\alpha \in H_d(D_{a,b})$. If $a > a_d$, Proposition 4.5(a) gives $\alpha = u^{a-a_d}\alpha'$ with $\alpha' \in H_d(D_{a_d,b})$. So we may assume $a \le a_d$. If $b > b_d$, then D_{a,b_d} is a branch because $b_d \ge 1$, and Proposition 4.5(b) gives $\alpha = v^{b-b_d}\alpha''$ with $\alpha'' \in H_d(D_{a,b_d})$. There are finitely many pairs (a,b) with $a \le a_d$ and $b \le b_d$, and each $H_d(D_{a,b})$ is a finitely generated abelian group by Lemma 4.4.

(b) By Remark 4.3, the map underlying u is homotopic to the stabilization map of Theorem 4.1 for $N = \mathbb{R}^{r+1}$. It induces an isomorphism $H_d(C_n^{+,\epsilon}(M)) \to H_d(C_{n+1}^{+,\epsilon}(M))$ for $n \ge 3d+5$. The groups with $2 \le n \le 3d+5$ are finitely generated by Lemma 4.4. □

Section 5 uses homology with coefficients in other finitely generated abelian groups.

Lemma 4.7. *Let Y be a space with commuting self-maps u and v such that $H_j(Y)$ is a finitely generated R-module for every j. Then $H_j(Y;A)$ is a finitely generated R-module for every j and every finitely generated abelian group A.*

Proof. The universal coefficient sequence

$$0 \to H_j(Y) \otimes A \to H_j(Y;A) \to \mathrm{Tor}\big(H_{j-1}(Y), A\big) \to 0$$

is natural in Y, hence a sequence of R-modules. Choose a presentation $0 \to \mathbb{Z}^k \to \mathbb{Z}^l \to A \to 0$. Then $H_j(Y) \otimes A$ is a quotient of $H_j(Y)^l$, and $\mathrm{Tor}(H_{j-1}(Y), A)$ is a submodule of $H_{j-1}(Y)^k$. Both are finitely generated because R is Noetherian. □

5 Bredon homology

5.1 Conventions

The constant Mackey functor $\underline{\mathbb{Z}}$ for the group C_2 takes the value $\mathbb{Z}$ on both orbits C_2/e and C_2/C_2; its restriction map is the identity and its transfer map is multiplication by 2 [1, Example 2.5.2]. Let $H\underline{\mathbb{Z}}$ be the associated Eilenberg–Mac Lane C_2-spectrum [6, Chapter XIII]. For a C_2-space Y we put

$$T_Y = H\underline{\mathbb{Z}} \wedge \Sigma^\infty_{C_2} Y_+, \qquad H_d^{C_2}(Y;\underline{\mathbb{Z}}) = \pi_d\big((T_Y)^{C_2}\big).$$

This is the Bredon homology of Y with coefficients in $\underline{\mathbb{Z}}$, evaluated at C_2/C_2; see [1, Section 2.1]. Smash products, fixed points and geometric fixed points of spectra are understood in the derived sense.

If Y is a C_2-CW complex, these groups are the homology of the cellular chain complex described in [1, Section 2.1]. For C_2 and $\underline{\mathbb{Z}}$ it is

$$C_*^{C_2}(Y;\underline{\mathbb{Z}}) = \frac{C_*(Y) \oplus C_*(Y^{C_2})}{\big(gc - c,\ \iota(c') - 2c'\big)}, \tag{5.1}$$

where C_* denotes cellular chains, c runs through $C_*(Y)$, c' runs through $C_*(Y^{C_2})$, and $\iota\colon C_*(Y^{C_2}) \to C_*(Y)$ is induced by the inclusion. The relation $gc = c$ comes from the automorphism of C_2/e, and

the relation $\iota(c') = 2c'$ from the projection $C_2/e \to C_2/C_2$ and the transfer of $\underline{\mathbb{Z}}$. If the action on Y is free, then (5.1) is the complex of coinvariants $C_*(Y)_{C_2} = C_*(Y/C_2)$, which gives

$$H_d^{C_2}(Y;\underline{\mathbb{Z}}) \cong H_d(Y/C_2;\mathbb{Z}) \qquad \text{for a free } C_2\text{-CW complex } Y. \tag{5.2}$$

Each $C_n^{+,\epsilon}(M)$ is a smooth manifold with a smooth involution, so $\mathcal{C}_{\mathrm{or}}(M)$ is a C_2-CW complex by Illman's triangulation theorem [5]. As explained at the end of Section 2.3, the groups obtained from $\mathcal{C}_{\mathrm{or}}(M)$ by the constructions of this section are R-modules and the natural maps between them are R-linear. Since R is Noetherian, submodules, quotients and extensions of finitely generated R-modules are finitely generated.

5.2 Geometric fixed points

For a C_2-spectrum T let $\Phi^{C_2}T = (\tilde{E}C_2 \wedge T)^{C_2}$ be its geometric fixed points, where $\tilde{E}C_2$ is the cofiber of $EC_{2+} \to S^0$.

Proposition 5.1. *For every $d \geq 0$, the R-module $\pi_d\Phi^{C_2}T_{\mathcal{C}_{\mathrm{or}}(M)}$ is finitely generated.*

Proof. Geometric fixed points commute with smash products and take the suspension spectrum of a C_2-space to the suspension spectrum of its fixed points [1, Propositions 2.2.2 and 2.2.3]. For a C_2-space Y this gives an equivalence

$$\Phi^{C_2}T_Y \simeq \Phi^{C_2}H\underline{\mathbb{Z}} \wedge \Sigma^\infty(Y^{C_2})_+,$$

natural in Y. By [1, Proposition 2.5.1], $\Phi^{C_2}H\underline{\mathbb{Z}}$ is a connective generalized Eilenberg–Mac Lane spectrum whose homotopy groups $A_i = \pi_i\Phi^{C_2}H\underline{\mathbb{Z}}$ are finitely generated. Fixing an equivalence $\Phi^{C_2}H\underline{\mathbb{Z}} \simeq \bigvee_{i\geq 0}\Sigma^i HA_i$, we obtain isomorphisms

$$\pi_d\Phi^{C_2}T_Y \cong \bigoplus_{i=0}^{d} H_{d-i}(Y^{C_2};A_i),$$

natural in Y. For $Y = \mathcal{C}_{\mathrm{or}}(M)$ each summand is a finitely generated R-module by Proposition 4.6(a) and Lemma 4.7. □

5.3 Homotopy orbits

For a C_2-CW complex Y let $Y_{hC_2} = EC_2 \times_{C_2} Y$, where EC_2 is a contractible free C_2-CW complex.

Lemma 5.2. *Let Y be a C_2-CW complex with commuting equivariant self-maps u and v such that $H_j(Y;\mathbb{Z})$ is a finitely generated R-module for every j. Then $H_d(Y_{hC_2};\mathbb{Z})$ is a finitely generated R-module for every d.*

Proof. The spectral sequence (4.1) for the free action of C_2 on $EC_2 \times Y$ has the form

$$E^2_{i,j} = H_i(C_2;H_j(Y;\mathbb{Z})) \Longrightarrow H_{i+j}(Y_{hC_2};\mathbb{Z}).$$

It is natural for equivariant maps, so it is a spectral sequence of R-modules. For a $\mathbb{Z}[C_2]$-module N, the groups $H_i(C_2;N)$ are the homology groups of the complex

$$\cdots \xrightarrow{1-g} N \xrightarrow{1+g} N \xrightarrow{1-g} N.$$

For $N = H_j(Y;\mathbb{Z})$ the maps $1 \pm g$ are R-linear, because u and v commute with g. So the groups $E^2_{i,j}$ are subquotients of a finitely generated R-module, and the same is true of the groups $E^\infty_{i,j}$. The spectral sequence lies in the first quadrant, so $H_d(Y_{hC_2};\mathbb{Z})$ has a finite filtration by R-submodules whose quotients are the groups $E^\infty_{i,d-i}$. □

5.4 Proofs of Theorem 1.1 and Corollary 1.2

Proof of Theorem 1.1. Let $Y = \mathcal{C}_{\mathrm{or}}(M)$ and $T = T_Y$. Smashing the cofiber sequence $EC_{2+} \to S^0 \to \tilde{E}C_2$ with T and taking fixed points gives the isotropy separation sequence

$$(EC_{2+} \wedge T)^{C_2} \longrightarrow T^{C_2} \longrightarrow \Phi^{C_2}T;$$

see [1, Lemma 2.4.1]. Since $EC_{2+} \wedge T = T_{EC_2 \times Y}$ and $EC_2 \times Y$ is a free C_2-CW complex with orbit space Y_{hC_2}, the homotopy groups of the first term are $H_*(Y_{hC_2}; \mathbb{Z})$ by (5.2). The long exact sequence of homotopy groups contains the exact sequence of R-modules

$$H_d(Y_{hC_2}; \mathbb{Z}) \longrightarrow H_d^{C_2}(Y; \underline{\mathbb{Z}}) \longrightarrow \pi_d \Phi^{C_2} T.$$

The left-hand term is finitely generated by Proposition 4.6(b) and Lemma 5.2, and the right-hand term by Proposition 5.1, so the middle term is finitely generated. □

Proof of Corollary 1.2. Let $R_0 = \mathbb{Z}[u, v^2]$. Then $R = R_0 \oplus vR_0$ is a finitely generated R_0-module, so $H_d^{C_2}(\mathcal{C}_{\mathrm{or}}(M); \underline{\mathbb{Z}})$ is finitely generated over R_0 by Theorem 1.1. The maps u and v^2 preserve the two sectors, and

$$H_d^{C_2}(\mathcal{C}_{\mathrm{or}}(M); \underline{\mathbb{Z}}) = H_d^{C_2}(\mathcal{C}^0_{\mathrm{or}}(M); \underline{\mathbb{Z}}) \oplus H_d^{C_2}(\mathcal{C}^1_{\mathrm{or}}(M); \underline{\mathbb{Z}})$$

as R_0-modules. A direct summand of a finitely generated module is finitely generated. □

6 Degree zero

6.1 Connected C_2-spaces

Lemma 6.1. *Let Y be a C_2-CW complex whose underlying space is path connected and whose fixed point space has finitely many path components $Y_1, \dots, Y_c$, with $c \geq 1$. Let $z_i \in H_0^{C_2}(Y; \underline{\mathbb{Z}})$ be the class of a point of Y_i. Then*

$$H_0^{C_2}(Y; \underline{\mathbb{Z}}) = \frac{\mathbb{Z}\langle z_1, \dots, z_c \rangle}{(2(z_i - z_j) : 1 \leq i < j \leq c)} \cong \mathbb{Z} \oplus (\mathbb{Z}/2)^{c-1}.$$

An equivariant map $f\colon Y \to Y'$ between two such complexes sends z_i to the class of the path component of Y'^{C_2} that contains $f(Y_i)$.

Proof. The quotient in (5.1) is the cokernel of a map of chain complexes. These complexes are concentrated in nonnegative degrees, so taking H_0 commutes with the cokernel:

$$H_0^{C_2}(Y; \underline{\mathbb{Z}}) = \frac{H_0(Y) \oplus H_0(Y^{C_2})}{\big(gc - c,\ \iota_*(c') - 2c'\big)}.$$

Here $H_0(Y)$ is infinite cyclic with a generator z_0 fixed by g, the group $H_0(Y^{C_2})$ is free on $z_1, \dots, z_c$, and $\iota_*(z_i) = z_0$. The relations are therefore $z_0 = 2z_i$ for $1 \leq i \leq c$. Eliminating z_0 leaves the relations $2z_i = 2z_j$, and in the basis $z_1, z_2 - z_1, \dots, z_c - z_1$ these say that the last $c-1$ basis elements have order two. The statement about maps follows from the description of z_i as the class of a point of Y_i. □

The proof of [1, Proposition 4.2.1] contains the same computation for unordered configuration spaces.

6.2 The presentation

Let $n \geq 2$ and $\epsilon \in \mathbb{Z}/2$. The space $C_n^{+,\epsilon}(M)$ is path connected, and by (3.1) its fixed point space is the union of the branches $D_{a,b}$ with $a + 2b = n$ and $b \equiv \epsilon \pmod 2$. There is at least one such branch, namely $D_{n,0}$ for $\epsilon = 0$ and $D_{n-2,1}$ for $\epsilon = 1$. We write

$$z_{a,b}\ (a \geq 2), \qquad z^+_{a,b},\ z^-_{a,b}\ (a \leq 1)$$

for the classes in $H_0^{C_2}(C^{+,\bar{b}}_{a+2b}(M); \underline{\mathbb{Z}})$ of the path components $D_{a,b}$ and $D^{\pm}_{a,b}$ of Theorem 3.3, and call them component classes. By Lemma 6.1, the group $H_0^{C_2}(C_n^{+,\epsilon}(M); \underline{\mathbb{Z}})$ is generated by the component classes of bidegree (n, ϵ), subject to the relations

$$2(z - z') = 0 \qquad \text{for component classes } z,\ z' \text{ of the same bidegree.} \tag{6.1}$$

Proposition 6.2. *The action of u and v on the component classes is given by*

$$\begin{aligned}
uz_{a,b} &= z_{a+1,b}, & vz_{a,b} &= z_{a,b+1} && (a \geq 2),\\
uz^{\pm}_{1,b} &= z_{2,b}, & vz^{\pm}_{1,b} &= z^{\pm}_{1,b+1} && (b \geq 1),\\
uz^{\pm}_{0,b} &= z^{\pm}_{1,b}, & vz^{\pm}_{0,b} &= z^{\pm}_{0,b+1} && (b \geq 1).
\end{aligned}$$

Proof. By Lemma 6.1 it suffices to determine the maps induced by u and v on path components of the fixed point spaces, and these can be read off from Proposition 3.5. □

In the notation of the introduction,

$$x = z_{2,0}, \qquad y = z^+_{0,1}, \qquad \delta = z^-_{0,1} - z^+_{0,1},$$

with $|x| = (2,0)$ and $|y| = |\delta| = (2,1)$.

Proof of Theorem 1.3. Write $H = H_0^{C_2}(\mathcal{C}_{\mathrm{or}}(M); \underline{\mathbb{Z}})$, and let P be the bigraded R-module with generators x, y, δ in the bidegrees above and relations

$$vx = u^2 y, \qquad 2(vy - u^2 x) = 0, \qquad 2\delta = 0, \qquad u^2\delta = 0. \tag{6.2}$$

Since δ does not occur in the first two relations and x, y do not occur in the last two, P is the direct sum in the statement of the theorem. We show that $P \cong H$.

The relations (6.2) hold in H. By Proposition 6.2, both vx and u^2y are equal to $z_{2,1}$, and u^2 sends $z^+_{0,1}$ and $z^-_{0,1}$ to the same class $z_{2,1}$, so $u^2\delta = 0$. The elements $vy - u^2x = z^+_{0,2} - z_{4,0}$ and δ are differences of component classes of the same bidegree, so $2(vy - u^2x) = 0$ and $2\delta = 0$ by (6.1). This gives an R-linear map $\varphi\colon P \to H$.

Proposition 6.2 also expresses every component class in terms of x, y and δ:

$$\begin{aligned}
z_{a,b} &= u^{a-2}v^b x && (a \geq 2),\\
z^+_{1,b} &= uv^{b-1}y, & z^-_{1,b} &= uv^{b-1}(y + \delta) && (b \geq 1),\\
z^+_{0,b} &= v^{b-1}y, & z^-_{0,b} &= v^{b-1}(y + \delta) && (b \geq 1).
\end{aligned} \tag{6.3}$$

Hence φ is surjective.

To construct the inverse, let ψ assign to each component class the element of P given by the right-hand side of (6.3). We check that ψ respects the relations (6.1). Put $\theta = vy - u^2x \in P$, so that $2\theta = 0$. From $vx = u^2y$ we get

$$v^2x = u^2vy = u^2\theta + u^4x. \tag{6.4}$$

The branches whose component classes have a given bidegree (n, ϵ) are the $D_{a,b}$ with $a + 2b = n$ and $b \equiv \epsilon \pmod 2$, so two consecutive ones are $D_{a,b}$ and $D_{a-4,b+2}$. By (6.4),

$$\psi(z_{a-4,b+2}) - \psi(z_{a,b}) = u^{a-6}v^b(v^2x - u^4x) = u^{a-4}v^b\theta \qquad (a \geq 6),$$

and directly from the definitions

$$\psi(z^+_{0,b+2}) - \psi(z_{4,b}) = v^b\theta, \qquad \psi(z^+_{1,b+2}) - \psi(z_{5,b}) = uv^b\theta.$$

Finally $\psi(z^-_{a,b}) - \psi(z^+_{a,b})$ is $v^{b-1}\delta$ for $a = 0$ and $uv^{b-1}\delta$ for $a = 1$. All these elements are annihilated by 2. Any two component classes of the same bidegree are connected by a chain of such steps, so ψ respects (6.1) and defines a homomorphism of abelian groups $\psi\colon H \to P$.

The map ψ is R-linear. Comparing Proposition 6.2 with (6.3), one sees that $\psi(uz) = u\psi(z)$ and $\psi(vz) = v\psi(z)$ for every component class z, with one case that needs an argument: $uz^{\pm}_{1,b} = z_{2,b}$. Here

$$u\,\psi(z^+_{1,b}) = u^2v^{b-1}y, \qquad u\,\psi(z^-_{1,b}) = u^2v^{b-1}(y + \delta),$$

and both are equal to $v^bx = \psi(z_{2,b})$ because $u^2y = vx$ and $u^2\delta = 0$. By construction $\varphi\psi$ is the identity on component classes, and $\psi\varphi$ is the identity on x, y and δ. So φ and ψ are inverse isomorphisms. □

By (6.3), $v^k\delta = z^-_{0,k+1} - z^+_{0,k+1}$ and $uv^k\delta = z^-_{1,k+1} - z^+_{1,k+1}$ for $k \geq 0$. In the summand $R/(2, u^2)\langle\delta\rangle$ these elements are nonzero, and $u \cdot uv^k\delta = 0$. This is the failure of injectivity of u described in the introduction.

Corollary 6.3. *For $n \leq 4$ the groups $H_0^{C_2}(C_n^{+,\epsilon}(M); \underline{\mathbb{Z}})$ are*

$$\begin{aligned}
(n,\epsilon) = (2,0):&\ \mathbb{Z}\langle x\rangle, & (2,1):&\ \mathbb{Z}\langle y\rangle \oplus \mathbb{Z}/2\langle\delta\rangle,\\
(3,0):&\ \mathbb{Z}\langle ux\rangle, & (3,1):&\ \mathbb{Z}\langle uy\rangle \oplus \mathbb{Z}/2\langle u\delta\rangle,\\
(4,0):&\ \mathbb{Z}\langle u^2x\rangle \oplus \mathbb{Z}/2\langle vy - u^2x\rangle \oplus \mathbb{Z}/2\langle v\delta\rangle, & (4,1):&\ \mathbb{Z}\langle vx\rangle.
\end{aligned}$$

Proof. By Example 3.4 the fixed point spaces have 1, 2, 1, 2, 3 and 1 path components in the six cases, and Lemma 6.1 gives the groups. The generators are obtained from (6.3); for $(4, 0)$ the component classes are $z_{4,0} = u^2x$, $z^+_{0,2} = vy$ and $z^-_{0,2} = vy + v\delta$. □

6.3 Comparison with unordered configurations

The map $\tilde{u}$ of Section 2.3 also induces equivariant maps $u\colon C_n(M) \to C_{n+1}(M)$. On Bredon homology, the maps

$$u_*\colon H_0^{C_2}(C_n(M); \underline{\mathbb{Z}}) \longrightarrow H_0^{C_2}(C_{n+1}(M); \underline{\mathbb{Z}})$$

are injective for every $n \geq 0$. Indeed, $C_n(M)$ is a path connected smooth C_2-manifold, hence a C_2-CW complex [5], and its fixed point space is the disjoint union of the spaces $I_{a,b} \cong C_a(F) \times C_b(B)$ with $a + 2b = n$ from the proof of Theorem 3.3, and each $I_{a,b}$ is path connected. Let w_b be the class of $I_{n-2b,b}$. By Lemma 6.1, $H_0^{C_2}(C_n(M); \underline{\mathbb{Z}})$ is the direct sum of the infinite cyclic group generated by w_0 and the cyclic groups of order two generated by $w_b - w_0$ for $1 \leq b \leq n/2$. The map u sends $I_{a,b}$ into $I_{a+1,b}$, so u_* sends the generators w_0 and $w_b - w_0$ to the generators with the same names in the corresponding decomposition of the target.

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